ARTHUR OUDEYER

ISAE-Supaero, Engineering Curriculum



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PROFILE

Currently in the engineering program at ISAE-Supaero (Toulouse), I am passionate about computer science and artificial intelligence, and their applications in aerospace and the modeling of physical and biological phenomena. I am aiming for a career in research or engineering in these fields.

EXPERIENCE

RITMO University of Oslo - *March to July 2026 - Research Internship*
Using similarities to living organisms as an optimization criterion for **quality-diversity algorithms** in the search for **robotic morphologies and movements**, supervised by *Kyrre Glette*
Artificial Intelligence - Novelty Search - Morphology & Motion

Sabena technics - *June-July 2025 - Technical Internship*
Technical Assistance on A300-600 ST

Valemo - *June 2021 - Observation*
Installation and management of **wind and solar farms**, control center, discovery of energy network management

Institut de Mécanique des Fluides de Toulouse - *June 2021 - Researcher Shadowing*
Flow control, boundary layer instabilities, discovery of IMFT and its **fluid mechanics** researchers

Dronisos - *July 2020 - Observation*
Development of **drone fleets** and creation of shows, synchronization testing, **simulation**

Aérocampus Aquitaine - *June 2019 - Observation*
Discovery of mechanical and avionics training programs

EDUCATIONAL

ISAE-SUPAERO *(Toulouse, France)*
2024-2026 - **General Engineering Program**
Fluid Mechanics – Computer Science – Aerodynamics
Embedded Systems – Control Systems – Mechanics
Maths - Thermodynamic - Electronics

Classe Préparatoire Lycée Saint-Louis *(Paris)*
2022-2024 - MP2I, PSI
Preparation for scientific entrance exams (**CPGE**)
Algebra – Analysis – Computer Science – Physics

Lycée Français Stanislas *(Montréal)* **- Terminale**
2021 - **Baccalaureate with Highest Honors** *(Mathematics, Physics, Computer Science)*

SKILLS

Technics :

- **Frameworks / Languages** : Python *(Advanced)*, Scikit-learn, C/C++, Java, RaspberryPI, Arduino, ESP32, Teensy, Pygame, PlatformIO, PyPlot, Linux, ROS
- **Tools / Software** : Git, Make, Docker, VScode, Pycharm, Ableton Live, Fusion 3D, OBS
- **Hands-on Skills :** 3D Printing, Electronics, Wood work, Laser cutting, model-making

Project Coordination :

- As President of the *Supaéro Robotics Club* (2025-26), I coordinated a team of 20 students for the complete design of autonomous robots and their participation in the French Robotics Cup.

Languages :

- French *(Native)*, English *(**TOEFL** - C1)*, Spanish *(B2)*

SCIENTIFIC INTERESTS

- **Sciences Clubs** :
  - Robotik ISAE-Supaero (President 2025-26), *French Robotic Cup* (2024-2025, 2025-2026)
  - StanRobotix (*First Robotics cup*) (Lycée Stanislas, 2021)
  - Club Maths+ (Lycée, 2020)
  - Club Technology (Autonomous Martian Rover, *contest CGénial*) (Collège, 2016-2019)
- **Participation :** Général contest in Mathematics (2022), Finalist Alkindi contest (Cryptologie, 2019)
- **Model Aircraft** : Design and construction, mechanics and electronics of RC models
- **Design of Educational Activity :** *(See **webpage**)*
  - Robots Activity for middle school and high school students
  - Design of a workshop on drone and drone programming for high school students

CULTURAL INTERESTS

- **Music :** music making on DAW
- **Workshop visual arts :** drawing, painting (2025)
- **Sport :** Tennis, Breakdance, Triathlon, Ultimate

Personal Projects

Computer science projects are an opportunity to explore, discover, and improve in many different fields.

Details : **Webpage**
Projects Videos Demo

IA for Science :

- Evolutionary Algorithms (**Cellular Automata**, Walking Creature via Machine Learning)
- Neural Networks (MLP) (**Prédictions de trajectoire d'avion en papier** à partir de vidéos)

Physics Engine :

- Newtonian Mechanics (Celestial Simulation, Springs)
- Verlet Integration (**2D Particle-Based Engine**)

Biology & Terrain :

- Behavior (Ants, Shoal of fish dynamic)
- Simulated-life (**Particle Life**)
- Terrain Generation (Map Generator)
- Gaz Simulation (Particle interaction)

Games :

- Arcade (see Space Frontier, Sky Fighter, FPS Jet, Mini Racer)
- Simulation (F1 Simulator engineer data simulation)
- Platformers (Platformer, MissionZ)

Images :

- Image Fourier Compression (DFT)
- 3D object renderer

Web :

- Live Tennis score web scrapper on RaspberryPI